

How to Generate a Signature Certificate

1. Boot 签名证书

- (1) 打开命令控制台 ,通过指令进入 boot tools 目录

```
D:\git\DS10\tool\signed_tool>cd about-tools-2022.03.01-win-x64
D:\git\DS10\tool\signed_tool\about-tools-2022.03.01-win-x64>
```

- (2) 输入 arelease.exe -c . -k --key-alg=rsa,将生成一组新的 boot 签名证书

```
\git\DS10\tool\signed_tool>cd about-tools-2022.03.01-win-x64
\git\DS10\tool\signed_tool\about-tools-2022.03.01-win-x64>arelease.exe -c . -k --key-alg=rsa
rsing command line parameters ...
nished parsing command line parameters
TICE: Creating new key for 'Root Of Trust key'
TICE: Creating new key for 'Trusted World key'
TICE: Creating new key for 'Non Trusted World key'
TICE: Creating new key for 'SCP Firmware Content Certificate key'
TICE: Creating new key for 'SoC Firmware Content Certificate key'
TICE: Creating new key for 'Trusted OS Firmware Content Certificate key'
TICE: Creating new key for 'Non Trusted Firmware Content Certificate key'
TICE: Creating new key for 'Non Trusted OS Content Certificate key'
\git\DS10\tool\signed_tool\about-tools-2022.03.01-win-x64>
```

- (3) 将在工具 \about-tools-2022.03.01-win-x64\config\security\key 目录下
生成一组新的 boot 签名证书

电脑 > Data (D:) > git > DS10 > tool > signed_tool > aboot-tools-2022.03.01-win-x64 > config > security > key			
名称	修改日期	类型	大小
ecdsa_non_trusted_world_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_nt_fw_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_nt_os_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_root_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_scp_fw_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_soc_fw_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_tos_fw_key.pem	2025/9/29 17:35	PEM 文件	
ecdsa_trusted_world_key.pem	2025/9/29 17:35	PEM 文件	
rsa_non_trusted_world_key.pem	2026/6/2 13:06	PEM 文件	
rsa_nt_fw_key.pem	2026/6/2 13:06	PEM 文件	
rsa_nt_os_key.pem	2026/6/2 13:06	PEM 文件	
rsa_root_key.pem	2026/6/2 13:06	PEM 文件	
rsa_scp_fw_key.pem	2026/6/2 13:06	PEM 文件	
rsa_soc_fw_key.pem	2026/6/2 13:06	PEM 文件	
rsa_tos_fw_key.pem	2026/6/2 13:06	PEM 文件	
rsa_trusted_world_key.pem	2026/6/2 13:06	PEM 文件	

(4) 将新生成的文件名 rsa 开头的证书文件拷贝到工程目录下:

DS10\demo\ql-config\quec-project\aboot\config\security\key

替换目录下的文件。

2. APP 签名证书

(1) 进入 DS10\tool\signed_tool\app-sign-tool 目录

(2) 双击 gen_key.exe

```

D:\git\DS10\tool\signed_tool\app-sign-tool\gen_key.exe
. Key information:
N: EBF336FF9A2475E73F71A1F87A9ECC4BD6A2AF0C502DD6961CD6B4D5B2B0B8C717B7D366EA9B890549C4604ED15AEB7FAAD8B5FE853F405A6585
2349E7AC32E37618D56209934392A0C800AF73BAAE338923D57CC3B540F1F04074A8ECEEA4E033AB8F03E72E5F481DB556003AA960123E279C18CCC
7E482110ECD70AE7DD593C1234265BC951297D23DA9ED0E150FB29F4BBB508C944B2F76D5597445D9CBF6D354486ADC6A179C34780B111767D74F3F9
0447EFF368D2EBA84706E4269AE72BAAB622D19FBD1E00AEBE4EC94D3FB9F4DD1E9FD8CF1D58143EADDE6BB19768207B8CB81078B39361EF479989C3
4608F23D344DF674FCC591B088F8FE4C6AD3
E: 010001
D: 08F99EECA5E6AC8C416B0DD31922F7057315BF4FDA17B913321B08481DE2D99CEBB2B3E8A71D78A59A7AF88F32F20AF2206DA14968328529B26
2F6EC08BC034FEFC850A2E234333DA8B9DDBD7966B326F3007F6A252B2EF596494688E715C97510E6A7440DD9D031251AC8DCD8C71C74229A2E26E
F01947AFAE07353BB0D6CDE801C967AD40A298CCF0ED952AA21AB1443D2E1F921CE130ECB8B4F05E9F3FAA5FD90132642E5935E427ED0C8C855548A1
CBB6C4339D769C8501277D39FB8DCAD33F457A6F9CCEB2678B0D1B60A0774A46AC4D7F1E148836EC54CF9B92DE6948090407FCBD61C388E9CF993D21
9901C0A545C39984F8CEB45FB522C466C2E5
P: FF98F6B117D5B1AC399AC464099B780E9B604FC1337D1371BA9F555043A8D841AFD662D8928222CC04EE4588F6F231E417D9DA9B867811D4F0A7
A56A3B882A96CA8A00DE331BB1DE144F3017618F5419E00CCD1577AA05DF0F5886D8E576AE697B536129EA4DECD746CF6D5E351D1F6AFF6FAEE416
A6307AF408DD9BEB3CD
Q: EC5254B76C52F891BA01C702C55C73E9F63BC601B39C6E33B129CDEBE0AEF292BB8C26A8BCCA10989019A3FA4A13EECD84ED7DF8A657857F88E
B8866849155CD148956674992B1E03D8C1CC44FCFF4C4D0189851EA2AECB1971730F778FC28888778BEEC259E50C4E00E8E67F0A33DD4051E2650057
25A8CAD7E06A3EDC391F
DP: 36B8BA001A393F0BBF5D7A346CB623530542EE704FA4B49801BB81196912EA11CE22991102F4E79C6D8111D40C02B859BC1D812710F7FDE15811
40CE8A5F3704B36968FBE27A8A202391DF3B3886B1A0B14D287891216A5B5F45F126AB6290B1E4330A737A9AD4E3DB3E285D2603F129083D11F575
28D4F0DD3D75E12B69F5
DQ: C3E8F46225D1E47E0C5BC6FA5A1A1E4619CE7E75D38DF7AE6F1DB033ED4DDB369D57F06C8E258D608C85551516F8334A368AB395C08069E172
64233FA9C2BA388B3F76B44E8E1A11E24DB6C3B2610AA60412375350564B74CA4233086E1B3D08B4E2428AC7EF44261E46F55E77DCB1C99456FA9F
A9533BB04FFBBD464C8CD
QP: BBA53FCB1E2E3706376D7F6062FBB933498A68385164E3163AB13EB9511549E868E53C5CDEA017BFF6EED4FA00025611784C99F2C0531349796
6F665A4CF05FFCC4CA2D0B245A2E397521DF92530F2C2B753C3936E2BCAD87C654CA036ECF55117287CDD3011158C3939494AF1DDA73E06BA7C3753
1223F8EB36892AEFC9FA0
. Writing key to file... ok
+ Press Enter to exit this program.

```

(3) 工具将生成一组 APP 签名证书.将证书重命名

keyfile_private_key.pem 改为 app_private_key.pem

keyfile_public_key.pem 改为 app_public_key.pem

(4) 将两个文件拷贝到 DS10\demo\ql-cross-tool\sign 目录下,替换原来的文件

3. 运行 secboot.bat 脚本,打开签名功能